



ITEP 414 – SYSTEM ADMINISTRATION AND MAINTENANCE

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Section: BSIT 4E

PART I. COMPANY PROFILE

Company Name: TechNova Solutions Inc.

Nature of Business

TechNova Solutions Inc. is a newly established software development startup that designs, builds, and maintains web and mobile applications for small and medium-sized enterprises (SMEs) across the Philippines. The company offers custom software development, systems integration, and IT consulting services, positioning itself as an affordable and reliable technology partner for growing businesses.

Company Vision

To become a leading software development company in Region IV-A, recognized for delivering innovative, secure, and reliable digital solutions that empower small and medium enterprises to compete in a digital economy.

Company Mission

- To deliver high-quality, scalable, and secure software solutions tailored to client needs.
- To build and maintain a resilient, well-managed IT infrastructure that supports business continuity.
- To foster a collaborative, innovative, and growth-oriented workplace for employees.
- To uphold professionalism, data privacy, and ethical technology practices in every engagement.

Office Location

3rd Floor, Innovation Hub Building, National Highway, Barangay Poblacion II, Santa Cruz, Laguna, Philippines 4009.

Organizational Structure

TechNova Solutions Inc. follows a flat, functional organizational structure appropriate for a 20-person startup. The Chief Executive Officer (CEO) oversees four department heads, each managing a small team. This structure keeps communication direct and decision-making fast while the company scales.

- Chief Executive Officer (CEO) — overall business direction and operations
- IT Manager / Junior System Administrator — leads the IT Department (software development, infrastructure, and technical support)
- HR Manager — leads the Human Resources Department (recruitment, employee relations, policy compliance)
- Finance Manager — leads the Finance Department (budgeting, payroll, accounting)
- Sales Manager — leads the Sales Department (client acquisition, account management, business development)

Employee Distribution



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DEPARTMENT	NO. OF EMPLOYEES	PRIMARY FUNCTION
Information Technology	5	Software development, network administration and technical support.
Human Resources	4	Recruitment, employee welfare and policy enforcement.
Finance	5	Budgeting, payroll and financial reporting.
Sales	6	Client relations, business development, and marketing coordination.
TOTAL	20	

PART II. ENTERPRISE HARDWARE INVENTORY

The following hardware inventory is meant to be used by all four departments on a day-to-day basis. All non-IT staffs (HR, Finance, Sales) are provided with standard desktop PCs because their work is mostly office-based and application-focused, whereas IT staffs are provided with laptops to ensure flexibility of work, visit to clients and remote debugging. Shared infrastructure is dimensioned with a certain buffer for the company's future expansion.

ASSET ID	HARDWARE	QTY.	DEPARTMENT	PURPOSE
HW-001	Desktop Computer	15	HR, Finance, Sales	Standard workstation for office productivity applications, accounting software, and CRM tools.
HW-002	Laptop	5	Information Technology	Portable workstation for software development, testing, and client-site technical support.
HW-003	Server (Tower/Rack)	1	Information Technology	Hosts internal applications, file sharing, version control services, and centralized authentication.
HW-004	Router	1	Information Technology	Routes traffic between the internal network and the ISP connection.
HW-005	Switch	2	Information Technology	Provides wired connectivity for 20+ endpoints across the office floor with room for expansion.
HW-006	Printer (Network)	2	All Departments	Shared network printing for documents, contracts, and reports; one unit per office wing.
HW-007	UPS (Uninterruptible Power Supply)	3	Information Technology	Protects the server, networking equipment, and critical workstations from power interruptions.
HW-008	Wireless Access Point	2	All Departments	Extends reliable Wi-Fi coverage across the single office floor



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				for laptops and mobile devices.
HW-009	NAS Storage	1	Information Technology	Centralized, redundant storage for shared files and local backups.
HW-010	External Backup Drive	2	Information Technology	Portable offsite/rotating backup of critical business and server data.
HW-011	Monitor	25	All Departments	One monitor per desktop/laptop user, with IT staff issued dual monitors for development efficiency.

PART III. ENTERPRISE SOFTWARE INVENTORY

All the software layers below will help employees with everyday productivity and equip the IT department with software tools to develop, virtualize, and secure the company's systems.

SOFTWARE	VERSION	LICENSE	PURPOSE
Windows 11 Pro	23H2	Volume License	Primary operating system for all desktops and laptops; supports domain join, BitLocker, and remote management.
Ubuntu Server	22.04 LTS	Free / Open Source	Operating system for the company server; hosts internal web applications, Git repositories, and file services.
Microsoft Office	365 Business Standard	Subscription	Word processing, spreadsheets, and presentations for HR, Finance, and Sales documentation.
VS Code	1.9x (latest)	Free / Open Source	Primary code editor for the development team.
Git	2.44	Free / Open Source	Version control for tracking and managing source code changes.
GitHub Desktop	3.x (latest)	Free	Graphical interface for managing GitHub repositories without command-line use.
VirtualBox	7.x	Free / Open Source (GPL)	Virtualization platform for testing applications in isolated environments before deployment.
Google Chrome	Latest	Free	Primary web browser for research, testing, and cloud-based tools.
Microsoft Defender	Built-in (Windows 11)	Included with OS	Baseline endpoint antivirus and threat protection for all workstations.
AnyDesk	Latest	Free / Business tier	Remote desktop access for IT helpdesk support and troubleshooting.
7-Zip	23.x	Free / Open Source	File compression and archiving for backups and file transfers.

PART IV. ENTERPRISE NETWORK INVENTORY

This inventory lists the physical networking components needed to connect all four departments to the internet and to each other, along with basic structured cabling supplies.

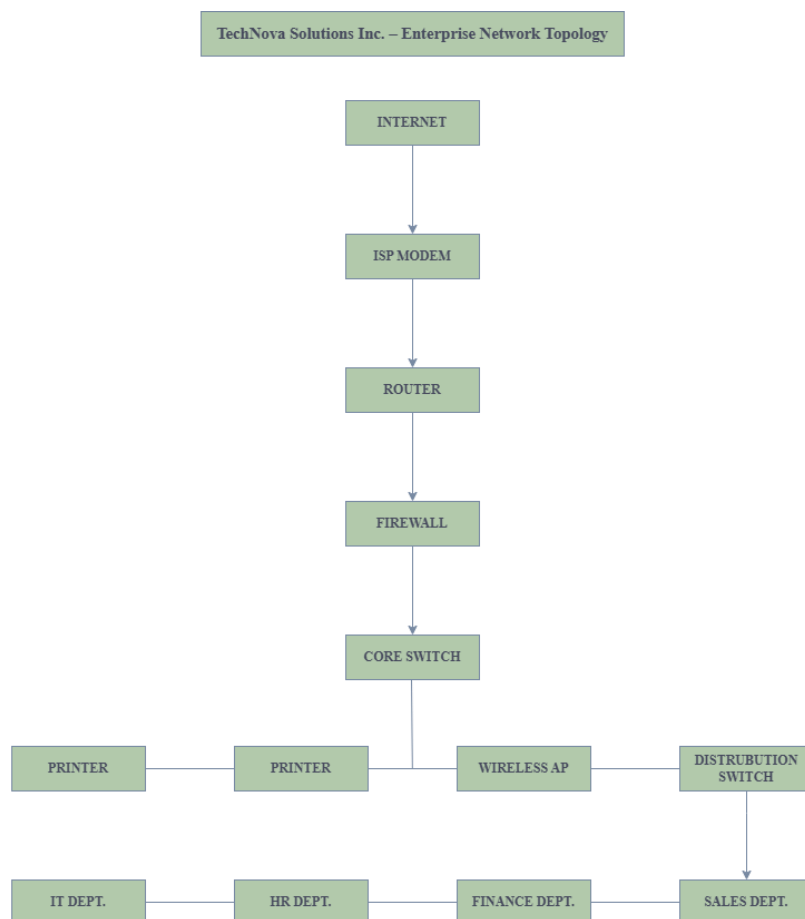
ITEM	QTY	SPECIFICATION	PURPOSE
ISP Modem	1	Fiber, business-grade	Provides the internet connection from the Internet Service Provider into the office.



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Router	1	Gigabit, dual-WAN capable	Routes traffic between the LAN and the internet; manages DHCP and NAT.
Firewall	1	Hardware/UTM appliance	Filters traffic, blocks intrusions, and enforces network security policies.
Switch	2	24-port Gigabit managed	Connects wired devices across departments and supports VLAN segmentation.
Access Point	2	Dual-band Wi-Fi 6	Provides wireless connectivity across the office floor.
Patch Panel	1	24-port CAT6	Organizes and terminates structured cabling runs to the switches.
CAT6 Cables	40	23 AWG, solid core	Wired connections between workstations, the patch panel, and network devices.

PART V. ENTERPRISE NETWORK DIAGRAM



PART VI. SYSTEM ADMINISTRATION ROLES

PART 6 — System Administration Roles

Helpdesk Technician

- **Responsibilities:**

- Supports employees with hardware, software and account problems in a first line technical support capacity.
- Logs, tracks and resolves tickets in a timely and agreed manner.



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- Carries out basic troubleshooting of desktops, laptops, printers and peripheral devices.
- Escalates complex issues to network/system administrators as necessary.
- **Skills:**
 - An entrepreneurial mindset with a strong entrepreneurial drive
 - Basic knowledge of Windows and Office troubleshooting
 - Tempered by the demands of time, patience and problem-solving ability has been developed.
 - Orderly student behavior in the classroom and during activities
- **Common Tools:**
 - Remotely supported tools (AnyDesk, TeamViewer)
 - Ticketing systems (such as Freshservice, Zendesk)
 - To help troubleshooting, the Windows Event Viewer and Device Manager information are provided.
- **Certifications:**
 - CompTIA A+
 - HDI Support Center Analyst (HDI-SCA)
 - Microsoft 365 Certified: Fundamentals

Network Administrator

- **Responsibilities:**
 - Installs, configures and troubleshoots the company's routers, switches and firewalls.
 - Tracks network performance, uptime and bandwidth utilizations.
 - Handles IP addressing, VLANs and Wi-Fi coverage.
 - Initiates network security protocols like access via VPN and firewall rules.
- **Skills:**
 - Familiarity with TCP/IP, Routing and Switching technologies
 - Understanding of Firewalls and VPNs
 - Skill in using a computer network for monitoring and troubleshooting
 - Documentation of topologies and change management.
- **Common Tools:**
 - Cisco Packet Tracer / GNS3
 - Wireshark
 - SolarWinds or PRTG Network Monitor
- **Certifications:**
 - Cisco Certified Network Associate (CCNA)
 - CompTIA Network+
 - Fortinet NSE



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Linux System Administrator

- **Responsibilities:**
 - Installs, configures and maintains Linux based servers.
 - Administers the users, file permissions and system services.
 - Do routine tasks automatically using shell scripting.
 - Installs security patches and checks system logs.
- **Skills:**
 - Knowledge of Linux commands and shell scripting.
 - Knowledge of file permissions, processes and services.
 - Package management and system hardening.
 - The rules for backing up and recovering data.
- **Common Tools:**
 - Bash/Shell scripting
 - SSH
 - cron
 - systemctl
 - Git
- **Certifications:**
 - CompTIA Linux+
 - Red Hat Certified System Administrator (RHCSA)
 - LPIC-1

Cloud Administrator

- **Responsibilities:**
 - Provisions and manages cloud-based resources.
 - Responsible for tracking cloud costs, performance and availability.
 - Applies identity and access management (IAM) policies.
 - Helps with cloud backup, disaster recovery, and scaling.
- **Skills:**
 - Experience with a large cloud platform
 - Knowledge of IAM, virtual networking and storage services
 - Optimization of costs and budget control over resources
 - Basics of Infrastructure as Code
- **Common Tools:**
 - AWS Management Console
 - Azure Portal
 - Terraform
 - CloudWatch / Azure Monitor
- **Certifications:**
 - AWS Certified Cloud Practitioner



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- AWS Certified SysOps Administrator
- Microsoft Certified: Azure Administrator Associate

HOW THESE FOUR ROLES WORK TOGETHER

These four positions within TechNova Solutions Inc. are a tiered hierarchy that ensures the seamless functioning of the organization's IT systems. The Helpdesk Technician acts as the initial support point for all day-to-day employee problems, and deals with simple issues promptly and appropriately, passing on anything beyond their capabilities. The Network Administrator is responsible for making sure the connectivity, routers, switches, firewalls, and Wi-Fi is stable and secure with no downtime, otherwise no one else would have anything to do. The Linux System Administrator provides maintenance to the company's internal server to ensure that services, permissions, and automation are all maintained to keep the server running and access to shared files and applications optimized. The Cloud Administrator is responsible for the same of the resources outside the office with respect to scaling, access from remote locations, and disaster recovery in the cloud. These roles all have a chain of responsibility to keep the company's IT infrastructure available, secure, and scalable: The helpdesk is tasked with protecting the end-user experience, the network administrator is responsible for protecting connectivity, the Linux administrator is responsible for protecting onsite resources, and the cloud administrator is responsible for protecting offsite resources.

PART VII: INFRASTRUCTURE RECOMMENDATION

Internet Provider

It is recommended that you use a business grade internet plan that supports a minimum of 100 Mbps symmetrical bandwidth such as PLDT Enterprise Fiber or Converge ICT Business Fiber services. The reason for a business plan is because it often has a Service Level Agreement (SLA) that guarantees uptime, quick support reaction time, and a static IP address, which is required for the business to host inside services and for enabling safe remote entry.

Server Specifications

At this size of the company, just one tower or 1U rack server is needed, and can be expanded if necessary. Intel Xeon E-2300 series processor, 32 GB ECC RAM, RAID 1 2x 1 TB SSD for redundancy, redundant or hot-swappable power supply. The RAID 1 and ECC RAM are recommended due to protection from data corruption and failure of any single drive: The server will be used for a 20-person company that currently has no backup system in place and the files they share, their internal applications and version control systems will be stored on the server.

Backup Strategy

In order to use the 3-2-1 backup method, TechNova needs to follow the rule of having 3 copies of data on 2 different types of media with 1 copy offsite. In reality the NAS device has a local backup of the server, the external backup drives are rotated weekly and stored off site and then there is a cloud backup service to store critical business data from a third location which is separate from the rest of the data. This layered approach means that the company is able to recover data if the office should suffer from fire, flood or theft.

Security Recommendations



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- Create VLANs by department, so that if one workstation is compromised there is limited lateral movement.
- Activate the intrusion prevention system (IPS) and turn off unused inbound ports on the hardware firewall.
- Use VPNs to access internal systems instead of services being exposed directly to the Internet.
- Follow the principle of least privilege: an employee needs access to the systems and files he or she will need during the course of his or her responsibilities.
- Regularly patch server, router, firewall and workstations.

Antivirus

For a startup company this size, the Microsoft Defender built into Windows 11 Pro is enough as an endpoint AV solution as it offers real-time protection, ransomware protection and centralized management through Microsoft 365, without the need to pay a separate license fee. When it comes to upgrading to a managed endpoint detection and response (EDR) solution, like Microsoft Defender for Business or Bitdefender GravityZone, is recommended, as it will help gain visibility of threats across all endpoints centrally.

Password Policy

- At least 12 characters (excluding spaces), including a mix of uppercase, lower case, numbers and symbols.
- Multi-factor authentication (MFA) for email, VPN and administrative accounts required.
- If a password is suspected to have been compromised, change it right away and do not use the same password on other systems.
- Allow only 5 attempts before account is locked out to prevent brute forcing.

Expansion Plan

- The current design (2 switches, 2 access points, 1 server) intentionally has headroom beyond the current 20 employees to support the growth to approximately 35-40 employees without a major redesign. As the company expands, TechNova should consider: Adding a second server for redundancy and load distribution, upgrading to 1 Gbps or higher business internet plan, scaling the capacity of the switches or adding a second network closet for growing office space, and even moving select workloads to a cloud provider to minimize reliance on on-prem hardware as user numbers increase.

PART VIII. PERSONAL REFLECTION

This Week 2 portfolio project helped me to gain understandings of IT infrastructure planning. Prior to this activity, I had mainly viewed networking, servers, security, and system administration as distinct areas. During the development of a full system for a fictional start up, I found that all these pieces must be connected as a system.

The most challenging part for me was creating the enterprise network diagram. It was simple enough to list the networking equipment, but it wasn't so easy to decide how everything should be connected. I would have to consider the positioning of the firewall and router, how



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the departments should be arranged and how the diagram should be interpreted. I also had to find the correct network symbols and know how the data would move around the network.

This project was also a learning experience for me on the need of proper planning before establishing a real system. I could end up spending a lot of money or not have enough equipment if I purchase without knowing the number of employees and what they need. For example, some workers might require a laptop computer because of their job requirements, and others might be better equipped with desktop computers. I also found out that one should be prepared for possible complications (such as server failures or problems with internet).

Overall, this project helped me a lot in developing my skills as an aspiring IT professional. I learned how to develop hardware and software inventories, draw network diagrams, identify IT roles and recommend infrastructure. Most importantly, I learned that decisions with regards to IT must be taken by carefully planning and reasoning. This will make me feel more confident about future system administration activities, and also give me a good understanding of the importance of planning before implementing something.